Cisco 7945G

Home telephony

Asterisk in k3s + Home Assistant

10 phones with color displays

Our lot: 10 Cisco 7945G phones

10

phones

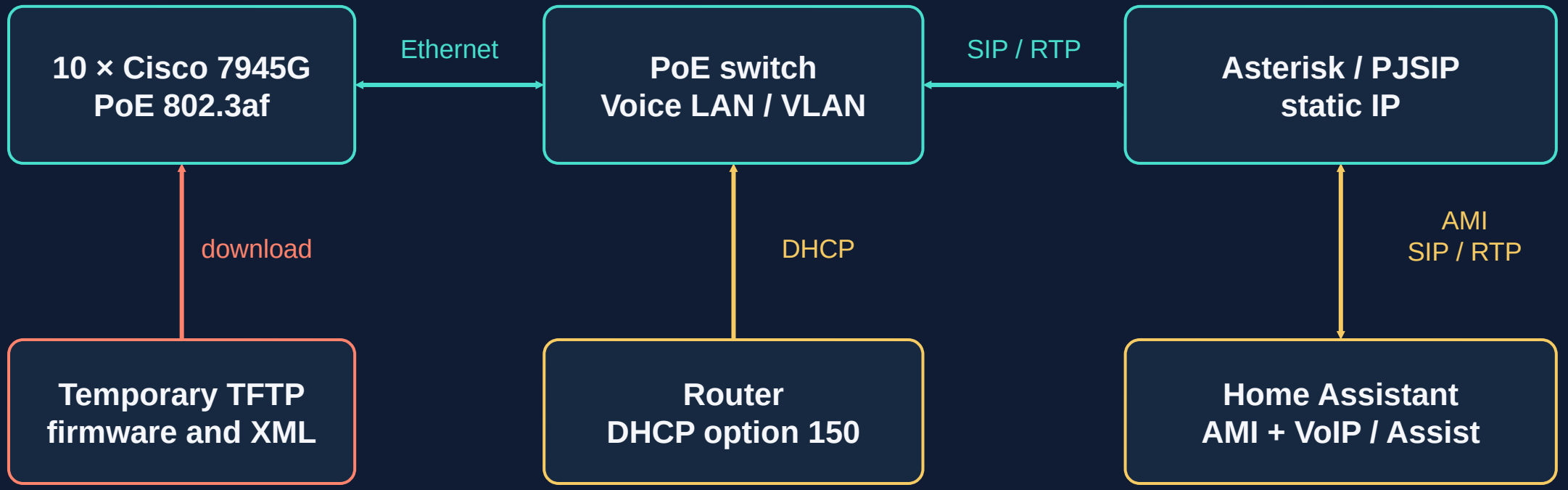
7945_7965 SIP firmware family

A phone running SCCP needs SIP firmware.
This model does not need MPP conversion.

Before starting

Record PID, VID/HW, MAC, App Load,
Boot Load, IP and TFTP for every phone.

Cisco, Asterisk and Home Assistant



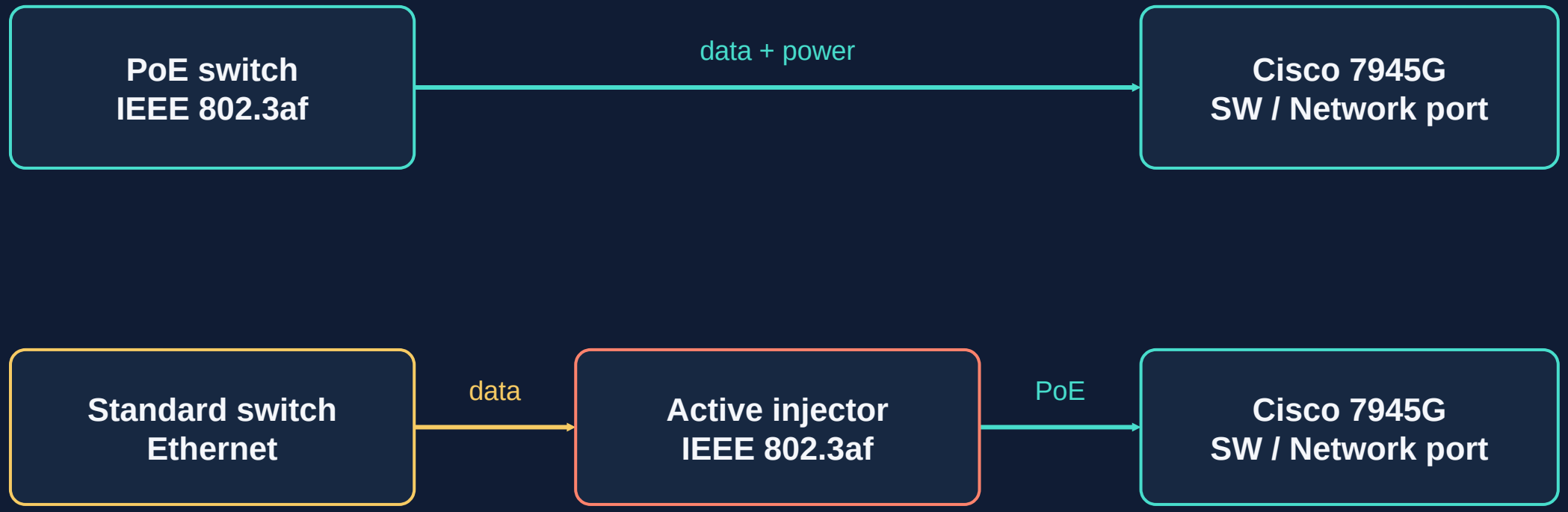
Asterisk uses hostNetwork and one replica. HA opens the AMI connection to the PBX.
TFTP runs only during provisioning. SIP, RTP and AMI stay on the LAN / VPN.

Signaling and audio ports

5060	TCP / UDP	SIP: Cisco TCP, HA UDP
10000–10199	UDP on PBX	RTP / audio
5038	TCP on PBX	AMI: HA source IP /32
69 + 20000–20031	UDP on PBX	Temporary TFTP

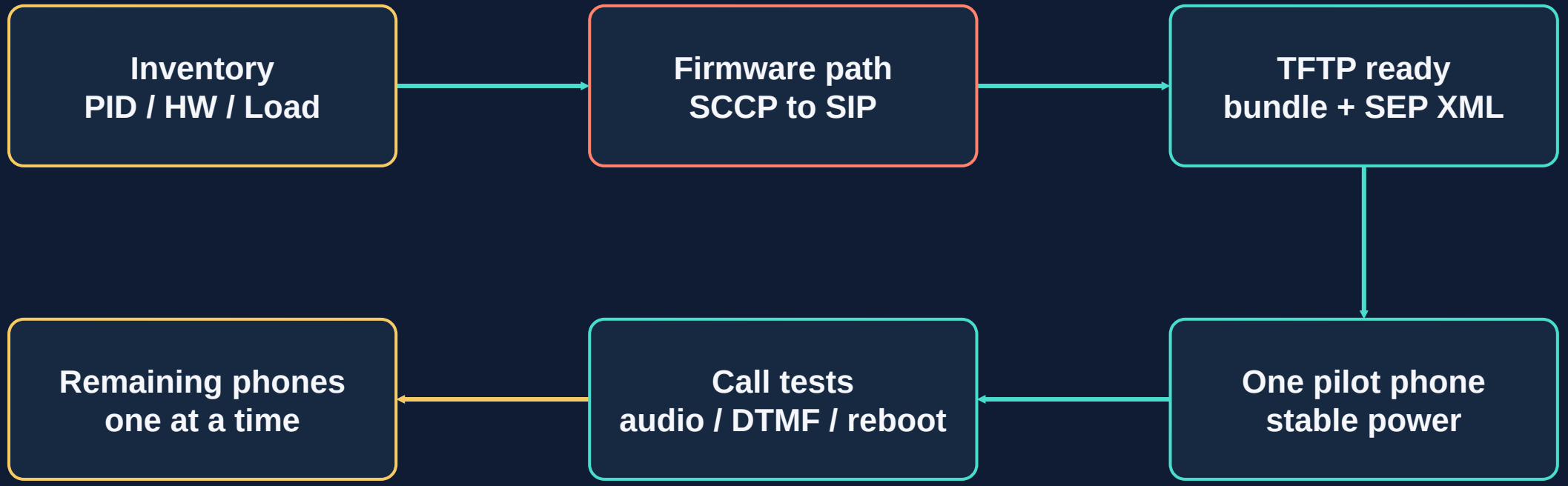
HA uses dynamic RTP / RTCP ports. Restrict firewall rules to the required source IPs.

7945G has built-in PoE support



A passive 24 V injector is unsuitable. The UPS also needs to power the switch, router and k3s node.

One pilot phone before the full fleet



HW 13.0+ requires at least 9.3(1)SR3. The upgrade path depends on the current load. Prepare TFTP and recovery before a reset. Keep power connected during flash writes.

TFTP preparation before restart

Complete official firmware bundle

.loads and every referenced binary file
with the original filenames and letter case

SEP<MAC>.cnf.xml

SIP server, extension and a unique password
Load ID from the selected 7945_7965 firmware branch

Use DHCP option 150 on the home router or Alternate TFTP on the phone.

Asterisk: one extension per phone

201 **202** **…210**

TCP with a separate endpoint, auth and AOR
A unique password for each phone

G.711 ulaw / alaw
direct_media=no keeps RTP on Asterisk

600 echo, 601 tone, 620 ring group. Extension 630 requires Assist.

AMI: status and call control

Home Assistant connects to PBX:5038

Status, connected line and DTMF
Actual entity IDs from your HA instance

Place calls and hang up a specific channel

asterisk.send_action: Originate / Hangup
The DTMF example sends a harmless notification

Use a separate secret and a /32 ACL. The system permission allows administrative actions.

630: voice assistant on the phone



First call: enable Allow calls and select a pipeline in HA.
Then test speech recognition, the spoken reply and hangup.

SIP From retains the phone extension. HA uses dynamic UDP ports for RTP / RTCP.
A call on the pilot phone verifies audio. Pod readiness checks server components.

Deployment and commissioning

01 **Server bootstrap**

Images, static IP, firewall, Asterisk and Opus

02 **Verified pilot phone**

PID / HW / Load / MAC, SIP profile, temporary TFTP

03 **Home Assistant setup**

AMI /32, secret, actual entity IDs and VoIP pipeline

04 **Acceptance and rollout**

Two phones, a call to 630, then the remaining phones

Acceptance tests on real phones

Telephony

201 and 202: registration, two-way audio, hold / transfer

Home Assistant

States, Originate, Hangup on the correct Channel, DTMF

Assist 630

Speech recognition, spoken reply and hangup

Repeat after phone, PBX and HA restarts. Readiness does not verify audio.

Operations and recovery

Private backups outside Git

Inventory, SIP / AMI secrets and Asterisk data
HA config / .storage, entity mappings and image versions

One node is a single point of failure

UPS for network equipment and the k3s node
A recovery plan that preserves the LAN IP

Exclude raw DTMF and connected-line data from Recorder. Review existing history separately.

Server readiness and phone acceptance

Server preparation can come first

Images, configuration, AMI and Opus
Tests verify the server components

A pilot verifies the phone experience

Phone inventory, a two-party call and Assist
Add the remaining phones after acceptance

Physical acceptance is a separate step. Keep the results in the private deployment log.